

NIFTY 100 FINANCIAL INTELLIGENCE PLATFORM

Comprehensive Platform Analyst & Developer Guide

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Target Audience:	Equity Research Analysts, Portfolio Managers, Developers
Backend API Server:	FastAPI (v1.0.0) — Port 8000
Web Application:	Streamlit Multi-Page App — Port 8501
Database Engine:	SQLite 3 (Indexed Relational Store)

Executive Notice: This technical guide provides step-by-step documentation for deploying, navigating, querying, and extending the Nifty 100 Financial Intelligence Platform. It includes REST API references, cURL commands, dashboard navigation workflows, PDF report generation instructions, data quality rule catalogs, and system troubleshooting protocols.

1. Project Overview & Architecture

The **Nifty 100 Financial Intelligence Platform** is an end-to-end quantitative equity research system designed for analyzing India's top 100 market-cap companies. The platform integrates an automated ETL data pipeline, relational SQLite storage, a multi-factor quantitative stock screener, financial ratio engines, clustering analytics, automated PDF report generation, and a high-performance FastAPI REST server.

Key Architectural Components:

Layer	Technologies Used	Functionality
Data Pipeline (ETL)	Python, Pandas, Regex	Normalizes dates/tickers, cleans raw Excel files, injects FK stubs.
Relational Database	SQLite 3, Composite B-Tree Indexes	Stores 10 financial tables with $O(\log N)$ query optimization.
Analytics Engine	NumPy, SciPy, Scikit-Learn	Computes CAGR, CFO Quality Score, K-Means clustering & peer ranks.
REST API Server	FastAPI, Uvicorn, Pydantic	Exposes 9 modular JSON endpoints with Swagger UI & OpenAPI docs.
Web Application	Streamlit, Plotly	8 interactive dashboard views with sidebar screening sliders.
Reporting Engine	ReportLab, PyPDF	Generates 1-page company tearsheets and 3-page portfolio PDFs.

2. Streamlit Dashboard User Guide

The Streamlit dashboard provides an intuitive, web-based portal for interacting with Nifty 100 financial data. Launch the dashboard by executing `streamlit run main.py`.

Navigating the 8 Dashboard Views:

Page / View	Key Features & Interactive Capabilities
01 Home / Overview	Market-wide summary KPIs, total market-cap distribution, top gainers/losers, and high-level sector breakdown charts.
02 Company Profile	Deep-dive company search with autocomplete. View 10-year P&L, Balance Sheet, Cash Flow, Valuation history, and peer radar charts.
03 Stock Screener	Interactive 10-slider quantitative screener with preset strategy buttons (Quality Compounders, Value, Growth, Debt-Free).
04 Peer Comparison	Side-by-side metric comparison, peer percentile rankings across 10 financial metrics, and radar chart overlays.
05 Financial Trends	Multi-year trend analysis for Revenue, Net Profit, OPM %, ROE %, and Debt-to-Equity across selected companies.
06 Sector Analytics	Broad sector median KPIs, index weight analysis, sector constituent tables, and cross-sector valuation comparisons.
07 Capital Allocation	Classifies companies into 7 capital allocation patterns (Reinvestor, Shareholder Returns, Distress, Liquidating) based on cash flow signs.
08 PDF Reports	On-demand PDF generation portal. Export 1-page company tearsheets or comprehensive 3-page portfolio summary PDF reports.

3. Quantitative Stock Screener Guide

The Stock Screener allows analysts to filter Nifty 100 companies using 10 quantitative financial metrics simultaneously. Filtering operates dynamically in real time.

Available Sidebar Filter Sliders:

Filter Name	Metric Field	Default Range	Financial Description
Return on Equity (ROE)	return_on_equity_pct	Min: 0% to 100%	Measures profitability generated per rupee of equity.
Debt to Equity (D/E)	debt_to_equity	Max: 0.0 to 10.0	Measures financial leverage (excl. Financials sector).
Free Cash Flow (Cr)	free_cash_flow_cr	Min: -5,000 to 50,000	Operating Cash Flow minus Capital Expenditures.
Revenue 5Yr CAGR	revenue_cagr_5yr	Min: -50% to 100%	Compounded annual growth rate of net sales over 5 years.
PAT 5Yr CAGR	pat_cagr_5yr	Min: -50% to 100%	Compounded annual growth rate of net profit over 5 years.
Price to Earnings (P/E)	pe_ratio	Max: 0 to 200	Valuation ratio comparing stock price to EPS.

Strategy Preset Buttons: Analysts can click one of 6 preset buttons in the sidebar to auto-populate slider configurations: *Quality Compounders*, *Growth Accelerators*, *Value Picks*, *Dividend Champions*, *Debt-Free Blue Chips*, and *Turnaround Candidates*.

4. Financial Ratios & KPI Methodologies

All financial indicators follow strict standard corporate finance definitions and edge-case handling rules.

Financial Metric	Mathematical Formula	Special Edge Case Handling Rules
Return on Equity (ROE)	$\frac{\text{Net Profit}}{\text{Equity Capital} + \text{Reserves}} \times 100$	Returns None if denominator ≤ 0 (negative/zero equity).
Debt to Equity (D/E)	$\frac{\text{Borrowings}}{\text{Equity Capital} + \text{Reserves}}$	Returns 0.0 for debt-free companies. Flags High Leverage if D/E > 5 (excl. Financials).
Interest Coverage Ratio	$\frac{\text{Operating Profit} + \text{Other Income}}{\text{Interest Expense}}$	Returns None & label 'Debt Free' if Interest = 0.
Compound Growth (CAGR)	$\left(\left(\frac{\text{End Value}}{\text{Beg Value}} \right)^{\frac{1}{n}} - 1 \right) \times 100$	Returns None with flag: 'TURNAROUND' (neg to pos), 'DECLINE_TO_LOSS' (pos to neg), or 'ZERO_BASE'.
CFO Quality Score	$\text{Average} \left(\frac{\text{CF O}}{\text{PAT}} \right) \text{ over 5 Years}$	Classifies score: High Quality (> 1.0), Moderate (0.5 to 1.0), Accrual Risk (< 0.5).

5. PDF Report & Tearsheet Generation

The platform includes a ReportLab PDF rendering engine capable of building publication-grade financial tearsheets and portfolio executive summaries.

1. Single-Company Tearsheets (1-Page PDF):

Contains company profile header, key valuation metrics, 5-year financial trend table, peer percentile radar chart, and automated NLP Pros & Cons bullet points.

Command / API Execution:

```
python scripts/generate_tearsheets.py --ticker TCS
```

Or via REST API: GET /api/v1/documents/tearsheet/TCS

2. Portfolio Summary Report (3-Page PDF):

Contains portfolio-wide KPI summary matrix, sector weight pie charts, 5-cluster profiling summary, distress alerts table, and deleveraging trends.

Command / API Execution:

```
python scripts/generate_portfolio_summary.py
```

Or via REST API: GET /api/v1/documents/portfolio-summary

6. REST API Developer Guide

The REST API is built on FastAPI and Uvicorn. All endpoints return standardized JSON payloads with strict Pydantic schema validation.

Base Server Configuration:

- Base URL: <http://127.0.0.1:8000>
- Common Route Prefix: `/api/v1`
- Interactive Swagger Docs: <http://127.0.0.1:8000/docs>
- ReDoc Specification: <http://127.0.0.1:8000/redoc>

Core API Routers & Endpoint Summary:

Router / Endpoint	HTTP Method	Description
<code>/api/v1/health</code>	GET	System status, uptime, DB connectivity, and 10 table row counts.
<code>/companies</code>	GET	List all 92 Nifty 100 constituent companies.
<code>/companies/{ticker}</code>	GET	Detailed profile, financial KPIs, and about text by ticker.
<code>/screener</code>	GET	Execute quantitative stock screener query across 7 filters.
<code>/sectors</code>	GET	List broad sectors with median ROE, PE, and Debt-to-Equity.
<code>/sectors/{sector}/companies</code>	GET	Get all companies and latest KPIs in a specific sector.
<code>/peers/{group_name}</code>	GET	Get 10 percentile ranks for companies in a peer group.
<code>/market-cap/{ticker}</code>	GET	Historical market cap, P/E, P/B, and EV/EBITDA time-series.
<code>/portfolio/stats</code>	GET	Portfolio-wide 10th, 25th, 50th, 75th, 90th percentile KPI stats.

7. cURL Code Examples & Request Samples

1. Health Check Request:

```
curl -X GET "http://127.0.0.1:8000/api/v1/health" -H "accept: application/json"
```

2. Stock Screener Query (Min ROE 15%, Max D/E 1.5):

```
curl -X GET "http://127.0.0.1:8000/api/v1/screener?min_roe=15&max_de=1.5" -H "accept: application/json"
```

3. Fetch Company Profile (TCS):

```
curl -X GET "http://127.0.0.1:8000/companies/TCS" -H "accept: application/json"
```

4. Fetch Sector Companies (Information Technology / IT):

```
curl -X GET "http://127.0.0.1:8000/sectors/IT/companies" -H "accept: application/json"
```

5. Fetch Peer Group Percentile Ranks:

```
curl -X GET "http://127.0.0.1:8000/api/v1/peers/IT%20Services" -H "accept: application/json"
```

6. Download Company PDF Tearsheet:

```
curl -X GET "http://127.0.0.1:8000/api/v1/documents/tearsheet/INFY" --output INFY_tearsheet.pdf
```


8. Data Quality Rules & Audit Framework

The platform enforces 14 automated Data Quality (DQ) validation rules split across CRITICAL and WARNING severity levels.

Rule ID	Severity	Rule Name & Condition Description
DQ-01	CRITICAL	Primary Key cannot be NULL or NaN.
DQ-02	CRITICAL	Duplicate Primary Keys across table records.
DQ-03	CRITICAL	Foreign Key violation (referencing non-existent company ID).
DQ-04	CRITICAL	Missing or empty financial year string.
DQ-05	CRITICAL	Missing mandatory schema fields.
DQ-06	CRITICAL	Sales / Revenue value is negative.
DQ-07	CRITICAL	Invalid ticker string pattern (must be uppercase alphanumeric).
DQ-08	CRITICAL	Duplicate (company_id, year) composite record.
DQ-09	WARNING	Operating Profit Margin (OPM) outside range [-100%, +100%].
DQ-10	WARNING	Balance Sheet mismatch (Assets $\neq$ Liabilities + Equity).
DQ-11	WARNING	Sales is equal to 0.0.
DQ-12	WARNING	Net Profit is negative (loss-making year).
DQ-13	WARNING	Total Assets value is negative.
DQ-14	WARNING	Financial year is in the future (> current year).

9. Troubleshooting & Operations Guide

1. Port Conflicts (Port 8000 or 8501 in Use):

If port 8000 or 8501 is already bound by another process on Windows:

```
netstat -ano | findstr :8000
taskkill /PID <PID> /F
```

2. Database Lock or Schema Errors:

If SQLite database throws `OperationalError: database is locked`, re-run the automated database loader script:

```
python scripts/etl/load_to_sqlite.py
```

3. Regenerating All Project Output Files:

To re-run the full analytics engine, screener presets, and report generation pipeline:

```
python scripts/run_presets_demo.py
python scripts/data_quality_audit.py
```

4. Running Automated Pytest Suite:

To run all 290+ unit, integration, and API tests with HTML coverage report:

```
pytest tests/ --html=reports/pytest_report.html -v
```